# Introduction to Networking and the OSI Model Theory

**Task 1: Wireshark Capture & Observed Protocol Names**

**How to Perform the Capture:**

1. Open **Wireshark** on your system.
2. Double-click your active connection (e.g., **Wi-Fi** or **Ethernet**) to start live packet capture.
3. Open a browser and visit **instagram.com**.
4. Return to Wireshark and click the red square button (**Stop Capture**).

**Protocols Observed in the Packet List:**

- **DNS (Domain Name System):** Resolves human-readable domain names (instagram.com) into destination IP addresses.
- **TCP (Transmission Control Protocol):** Manages connection handshakes (SYN, SYN-ACK, ACK) and reliable data transfer.
- **TLSv1.2 / TLSv1.3 (Transport Layer Security):** Secure, encrypted handshakes and encrypted application data streams.
- **QUIC / HTTP/3 (over UDP):** Modern web protocol used by Meta services for faster load times.
- **ARP (Address Resolution Protocol):** Maps local IP addresses to physical MAC addresses on the local network.
- **DHCP / MDNS:** Local network configuration and device discovery background traffic.

**Task 2: 3 Selected Packets & Corresponding OSI Layers**

| Selected Packet | Protocol Shown | Packet Info Details | Primary OSI Layer | Reason / Explanation |
|---|---|---|---|---|
| **Packet 1** | **DNS** | Standard query 0x1a2b A instagram.com | **Layer 7** | |

*(Application)* | Initiated directly by the browser to translate the domain name into an IP address before requesting web data. | | **Packet 2** | **TLSv1.3** | Application Data / Client Hello | **Layer 6**

*(Presentation)* | Handles cryptographic negotiation, TLS key exchange, and encryption/decryption of payload contents. | | **Packet 3** | **TCP** | 54210 → 443 [SYN] Seq=0 Win=64240 Len=0 | **Layer 4**

*(Transport)* | Operates using port numbers (443 for HTTPS) to establish a reliable end-to-end communication channel before data flows. |

**Task 3: Colorful OSI Model Diagram with Everyday App Examples**

The reference diagram below maps all 7 OSI layers to everyday apps (WhatsApp, Instagram, YouTube, and Google Maps):




**Layer-by-Layer App Summary:**

- **Layer 7 — Application: WhatsApp / Chrome / Zomato** →\rightarrow→ User-facing graphical interface and API calls where you type messages or browse feeds.
- **Layer 6 — Presentation: Instagram Images & Reels (JPEG / MP4)** →\rightarrow→ Formats, compresses media, and applies TLS 1.3 encryption to protect user privacy.
- **Layer 5 — Session: WhatsApp Audio/Video Calls (SIP)** →\rightarrow→ Establishes, coordinates, and cleanly closes real-time calling sessions between caller and receiver.
- **Layer 4 — Transport: YouTube Streaming & BGMI/Free Fire (TCP / UDP)** →\rightarrow→ Manages ports, reliable packet delivery (TCP for messages), and low-latency streams (UDP).
- **Layer 3 — Network: Google Maps / Router IP Routing** →\rightarrow→ Routes packets using logical IP addressing across internet routers between your phone and cloud servers.
- **Layer 2 — Data Link: Wi-Fi Access Point / Network Card** →\rightarrow→ Delivers frames using physical hardware MAC addresses to local router antennas.
- **Layer 1 — Physical: Fiber Optic Cables & RF Waves** →\rightarrow→ Carries raw binary bits as radio waves in the air or light pulses through fiber glass.

**Task 4: How a WhatsApp Message Travels Through 4 OSI Layers**

When you type *"Hey!"* and press **Send** in WhatsApp, the message travels through the following layers:

1. **Application Layer (Layer 7):**

You type the message inside the WhatsApp chat UI. When you tap Send, the app formats your text into an application-level message payload and prepares it for transmission.

2. **Presentation Layer (Layer 6):**

WhatsApp applies **End-to-End Encryption (E2EE)** using the Signal Protocol. Your plaintext message is scrambled into encrypted ciphertext so that no one (not even ISPs or Meta) can read it in transit.

3. **Transport Layer (Layer 4):**

The encrypted payload is packaged into a **TCP segment** with destination port **443** (or 5222). TCP attaches sequence numbers to ensure reliable, in-order delivery and automatic retransmission if packets drop over a weak connection.

4. **Network Layer (Layer 3):**

The TCP segment is wrapped inside an **IPv4/IPv6 packet**. Your smartphone's IP address is attached as the source, and WhatsApp's cloud server IP is attached as the destination, enabling internet routers to forward the message toward the recipient.